

Dx & Screening

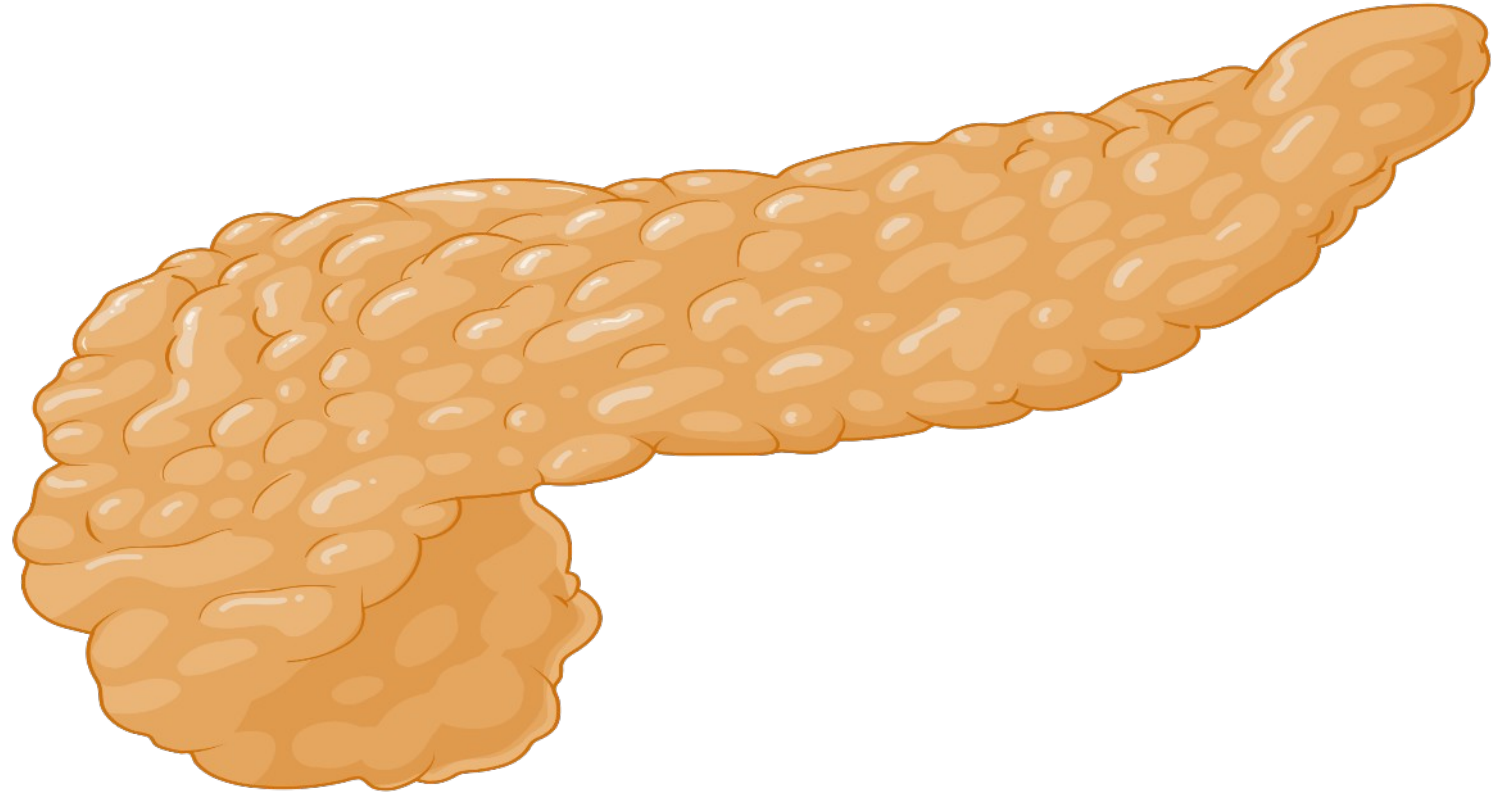
Complications

Management

Interventions

Cases

Type II Diabetes



Dx & Screening

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Diagnostic Criteria?

Symptoms?

no

yes



Weight
loss



Polydipsia



Polyuria



Blurry
vision

Fasting BG > 125 mg/dL
HbA1C > 6.5%
2 occasions or tests

Random BG > 200 mg/dL

Who should be screened?



BMI > 25 kg/m²
+ risk factors



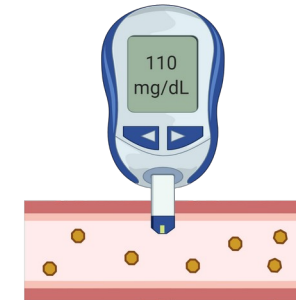
Age > 35



History of
gestational DM



Chronic steroids,
antipsychotics &
HIV meds



History of Pre-diabetes
(HbA1C > 5.7%)

Q3

How frequently to screen?

Q1 year

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COMPLICATIONS

MACROvascular disease

Cerebrovascular disease

CAD

PAD

Diabetes damages blood vessels

microvascular disease

Retinopathy

Nephropathy

Peripheral neuropathy

Retinal Scan

Urine
microalbumin / Cr
(>30mg/g)

1. Gross assessment for ulcers and nail care
2. Monofilament testing

Screening

Evaluate based on symptoms

Prevention

Treat vascular risk factors aggressively!

- BP control
- Tobacco cessation
- Lipid management

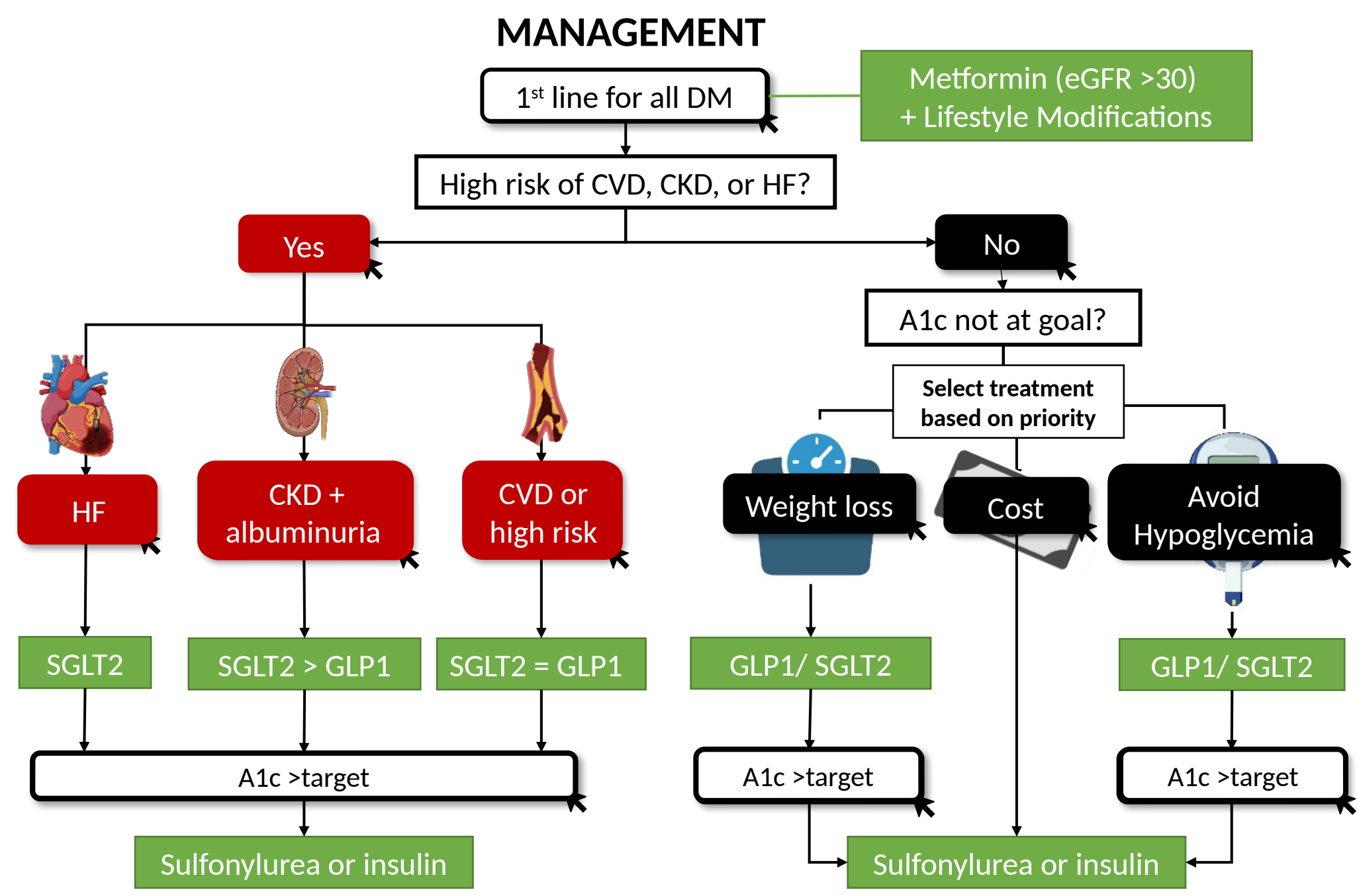
Screening

Yearly!

Prevention

Aggressive HbA1c target of <7%

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	Cost	PO/SQ	Wt	Cardiac		Renal	Side Effects	A1c
				HF	CVD			
Metformin	\$	PO	↔	-	?	-	GI upset Avoid in GFR<30	↓ ↓
Liraglutide, dulaglutide, semaglutide	\$\$\$	SQ/PO*	↓	-			Thyroid ca Pancreatitis GI upset	↓ ↓
Canagliflozin, SGLT2 inhibitors, empagliflozin	\$\$\$	PO	↓				Hypotension DKA Amputations Fournier's, UTIs	↓
Glipizide, glyburide, Sulfonylurea, glimepiride	\$	PO	↑	-	-	-	Hypoglycemia No glyburide in CKD	↓ ↓
Glargine, NPH, Insulin, Lispro, Aspart	\$	SQ	↑	-	-	-	Hypoglycemia	↓ ↓ ↓
Diet	\$	-	↓	-	-	-	High relapse rates	↓
Surgery	\$\$\$	-	↓	-	?	-	High relapse rates	↓ ↓

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1-3

4-6

Case 1

Would you screen the following asymptomatic patients for diabetes?

23 yo G1P1 F with a history of gestational DM

Yes. She will need lifelong screening at least every 3 y.

33 yo Asian F with BMI of 23 with a history of PCOS

Yes. Asian Americans with BMI > 23 (vs. 25 for other ethnicities) with RFs should be screened.

28 yo with schizophrenia on olanzapine . BMI 22.

Yes. Patients on certain medications (thiazides, steroids, HIV meds, atypical antipsychotics should be considered.

HbA1c is 6.4%. When should they be screened again?

Patients with borderline results should have a repeat test in **3-6 months**.

Case 2

55 yo presents to clinic to establish care. Reports 2-3 months of polydipsia and polyuria. BMI is 42.

What additional tests would you order?

BMP, Urine albumin/Cr, lipid panel.
Retina + foot exam.

What referrals are indicated?

Nutrition, exercise, and **weight loss** counseling. Then consider **surgery** (BMI > 40).

What medications would you recommend?

Metformin and **insulin** (A1c >10%).
At follow-up, add **GLP1** (obesity).

What if they wanted to avoid needles?

Metformin, **SGLT2**, oral **GLP1** (semaglutide), +/- **sulfonylurea**

Case 3

42 yo with a history of SLE, ESRD on HD, hypertension, hyperlipidemia presents to establish care.

Meds: prednisone, erythropoietin, amlodipine, atorvastatin.

Exam: BMI 22. BP 115/74.

Labs: Fasting BG 160, HbA1c 5.6%.

Does this patient have diabetes?

Very likely. Confirm with a second fasting BG > 120. The **A1c is discrepant** and underestimates average BG with high RBC turnover (erythropoietin).

You confirm diagnosis of T2DM. What medications would you recommend?

Reduced dose insulin. Metformin is contraindicated with GFR <30. Few data on SGLT2, GLP1 and oral agents in ESRD.

What is your glycemic target?

A1c 6.5-7.5% in ESRD. Fasting BG/ CGM can be used when A1c is inaccurate.

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Case 4

45 yo with T2DM and tobacco use presents for routine DM follow up. Meds include metformin 2000 mg ER and simvastatin.
Exam: BP 152/83, foot exam nml, BMI 24.
Labs: A1c 7.4%, Cr 1.2, Urine albumin:Cr ratio is 303 mg/g

How would you treat the hypertension?

ACE inhibitor or **ARB** (given albuminuria)
+ lifestyle modifications

What meds would you use to treat DM?

SGLT2 to improve renal outcomes given diabetic nephropathy with albuminuria.

What nonpharmacologic intervention would lower ASCVD risk?

Tobacco cessation counseling

Their A1c ↑ after losing their job. What is the next step in management?

Assess social determinants of health and mental health.

Case 5

82 yo with T2DM, COPD, HTN, CAD, diabetic CKD stage 2, BPH, and cognitive impairment presents for recurrent falls. A1c is 7.5%.
Meds: metformin 1000 mg BID, glargine 10 U qHS, lisinopril 40 mg, tamsulosin, atorvastatin 40 mg.

What is your A1c goal?

<8% given age and comorbidities, less stringent if poor overall prognosis

How will you adjust their meds?

Discontinue insulin. Consider starting **SGLT2** for diabetic CKD.

...One year later, eGFR ↓ to 20. A1c is 8.3%.

How will you adjust their meds?

Stop **metformin** (GFR<30). Given worsening comorbidities, accept A1c up to 8.5%.

Case 6

How would you manage the following patients' diabetes therapies?

45 yo with T2DM on metformin, PAD s/p amputation. A1c 6.5%.

GLP1 for ASCVD regardless of A1c. Avoid SGLT2 (amputations).

53 yo homeless and uninsured, T2DM on metformin. CKD3. A1c 8.5%.

Glipizide (not glyburide given CKD). Insulin may be hard to manage.

72 yo with T2DM on metformin, prior DKA, HFrEF. A1c 8.6%.

GLP1 given HFrEF. Avoid SGLT2 (DKA).

67 yo with CKD3, T2DM on metformin and SGLT2 with a femur fracture after a ground level fall. A1c 8.5%

Stop SGLT2 (possible ↑ risk of fracture). **Start GLP1** (renoprotective)

Practice Update — ADA Standards of Care 2025–2026

- **Organ-protection drugs are now a CORE strategy, independent of A1c or metformin. Start an SGLT2 inhibitor and/or GLP-1 RA with proven benefit in anyone with ASCVD (or high risk), HF, or CKD — even at A1c goal.**
- Metformin is no longer mandated first for everyone; comorbidity-driven agents may be started first (metformin still reasonable, low-cost, eGFR ≥ 30).
- Consider INITIAL combination therapy when A1c is 1.5–2.0% above goal, or with established/high-risk CVD.
- Weight + glycemia: prefer a GLP-1 RA or dual GIP/GLP-1 RA (semaglutide, tirzepatide) for greater weight loss; useful in obesity, MASLD/MASH, HFpEF.
- HF (HFrEF and HFpEF): SGLT2 inhibitor recommended for glycemic control + fewer HF hospitalizations.
- CKD with albuminuria despite ACEi/ARB + SGLT2i: add finerenone (nonsteroidal MRA) to further reduce CKD progression and CV events.
- Screening unchanged: start at age 35 (or earlier with overweight/obesity + ≥ 1 risk factor).

Source: ADA Standards of Care in Diabetes 2025 & 2026 (Diabetes Care, Suppl. 1, Sections 2, 9, 10, 11).

Take-Home Points

- Screen for T2DM at age 35, or earlier if BMI ≥ 25 (≥ 23 in Asian Americans) plus a risk factor; every 3 yr, or yearly if prediabetes (A1c 5.7–6.4%).
- Diabetes drives microvascular (retinopathy, nephropathy, neuropathy) and macrovascular (ASCVD) disease. Screen microvascular complications yearly; tight A1c ($< 7\%$) prevents micro- but not established macrovascular disease.
- Individualize therapy by comorbidity FIRST, then A1c. ASCVD/HF/CKD $\rightarrow$ SGLT2i and/or GLP-1 RA regardless of A1c (mortality/renal benefit; insulin has none).
- Liberalize A1c to 8–8.5% in older adults with limited life expectancy or many comorbidities; avoid hypoglycemia.

Attribution & Optimization Notes

- Adapted from TeachIM.org "Outpatient Management of Type II Diabetes Mellitus" (Brett, Guinn, Melton, Trefilek, Fainstad; exec. ed. Zhang). Published Mar 2022, last reviewed Jun 2023.
- teachim.org/teaching_material/op-dm2/ — Licensed CC BY-NC 4.0 (attribution, non-commercial). Educational use only; clinician-reviewed.
- Surgical corrections to original deck: "duraglitide" → "dulaglutide" (slide 5); "SGTL2" → "SGLT2" ×2 (slide 4); UACR "303 mg/dL" → "303 mg/g" (case 4, slide 7).
- Added (source-cited): ADA 2025–2026 Standards of Care update slide (organ-protection as core therapy, initial combination therapy, GIP/GLP-1 dual agonists, finerenone for residual-albuminuria CKD).
- Companion files in package: Facilitator Guide (T2DM 2023), Learner Handout — Treatment Options, medication-class table.
- Optimization run: 2026-06-10. Medical content should be clinician-reviewed before teaching.